LAB 7 SE

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Objective

To explore fuzz testing as a security and quality-assurance technique by using the Hypothesis library to identify input-validation bugs in the Pet Walking Service program and improve the robustness of its functions.

Software and Tools

• Python 3.13.7

• Hypothesis 6.170.15

• Pytest 9.2.0

• Visual Studio Code / Windows PowerShell

• Python virtual environment (venv)

Functions Tested

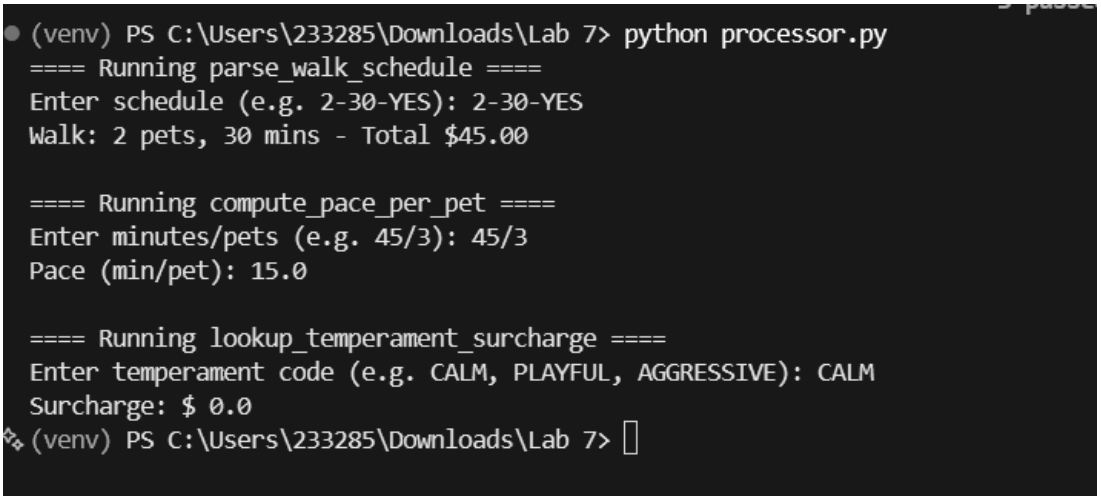
• **parse_walk_schedule()**: parses pet count, walk duration and weekend status, then calculates the walk fee.

• **compute_pace_per_pet()**: calculates walking minutes per pet.

• **lookup_temperament_surcharge()**: returns the surcharge associated with a pet temperament.

Procedure

The required packages were installed in a virtual environment using `pip install hypothesis pytest`. The application was executed with valid sample inputs to verify its normal output.



Three Hypothesis tests were written with `@given(st.text() | st.none())`. Each test passes generated strings and None values to one target function and fails if an exception is raised.

The first fuzz-test run exposed invalid-input handling problems in all three functions. Generated examples included empty strings, None and malformed input formats.

Fuzz-Test Failures



```
(venv) PS C:\Users\233285\Downloads\Lab 7> python -m pytest test_processor.py
============================= test session starts =============================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\233285\Downloads\Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_processor.py FFF                                                    [100%]

================================== FAILURES ===================================
_____________________________ test_parse_walk_schedule_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 12, in test_parse_walk_schedule_no_crash
  |     def test_parse_walk_schedule_no_crash(s):
  |                   ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 14, in test_parse_walk_schedule_no_crash
    |     parse_walk_schedule(s)
    |     ~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\233285\Downloads\Lab 7\processor.py", line 2, in parse_walk_schedule
    |     pets_str, mins_str, weekend_str = schedule_spec.strip().split('-')
    |     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 3, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 16, in test_parse_walk_schedule_no_crash
    |     assert False, f"parse_walk_schedule crashed with input {s!r}: {e}"
    | AssertionError: parse_walk_schedule crashed with input '': not enough values to unpack (expected 3, got 1)
    | assert False
    | Failing test case: test_parse_walk_schedule_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 14, in test_parse_walk_schedule_no_crash
    |     parse_walk_schedule(s)
    |     ~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\233285\Downloads\Lab 7\processor.py", line 2, in parse_walk_schedule
    |     pets_str, mins_str, weekend_str = schedule_spec.strip().split('-')
    |                                       ^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 16, in test_parse_walk_schedule_no_crash
    |     assert False, f"parse_walk_schedule crashed with input {s!r}: {e}"
    | AssertionError: parse_walk_schedule crashed with input None: 'NoneType' object has no attribute 'strip'
    | assert False
    | Failing test case: test_parse_walk_schedule_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________ test_compute_pace_per_pet_safe _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 21, in test_compute_pace_per_pet_safe
  |     def test_compute_pace_per_pet_safe(s):
  |                   ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 23, in test_compute_pace_per_pet_safe
    |     compute_pace_per_pet(s)
    |     ~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\233285\Downloads\Lab 7\processor.py", line 13, in compute_pace_per_pet
    |     mins = float(mins_str)
```

```
_____ test_compute_pace_per_pet_safe _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 21, in test_compute_pace_per_pet_safe
  |     def test_compute_pace_per_pet_safe(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 23, in test_compute_pace_per_pet_safe
    |     compute_pace_per_pet(s)
    |     ~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\233285\Downloads\Lab 7\processor.py", line 13, in compute_pace_per_pet
    |     mins = float(mins_str)
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 25, in test_compute_pace_per_pet_safe
    |     assert False, f"compute_pace_per_pet crashed with input {s!r}: {e}"
    | AssertionError: compute_pace_per_pet crashed with input '/': could not convert string to float: ''
    | assert False
    | Failing test case: test_compute_pace_per_pet_safe(
    |     s='/',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 23, in test_compute_pace_per_pet_safe
    |     compute_pace_per_pet(s)
    |     ~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\233285\Downloads\Lab 7\processor.py", line 12, in compute_pace_per_pet
    |     mins_str, pets_str = walk_metrics_str.split('/')
    |     ^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 25, in test_compute_pace_per_pet_safe
    |     assert False, f"compute_pace_per_pet crashed with input {s!r}: {e}"
    | AssertionError: compute_pace_per_pet crashed with input '': not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_compute_pace_per_pet_safe(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 23, in test_compute_pace_per_pet_safe
    |     compute_pace_per_pet(s)
    |     ~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\233285\Downloads\Lab 7\processor.py", line 12, in compute_pace_per_pet
    |     mins_str, pets_str = walk_metrics_str.split('/')
    |                          ^^^^^^^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 25, in test_compute_pace_per_pet_safe
    |     assert False, f"compute_pace_per_pet crashed with input {s!r}: {e}"
    | AssertionError: compute_pace_per_pet crashed with input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_compute_pace_per_pet_safe(
    |     s=None,
    | )
    +------------------------------------
```

```
_____ test_lookup_temperament_surcharge_safe _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 30, in test_lookup_temperament_surcharge_safe
  |     def test_lookup_temperament_surcharge_safe(s):
  |                  ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 32, in test_lookup_temperament_surcharge_safe
    |     lookup_temperament_surcharge(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\233285\Downloads\Lab 7\processor.py", line 26, in lookup_temperament_surcharge
    |     return surcharges[prefix]
    |            ~~~~~~~~~~^^^^^^^^
    | KeyError: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 34, in test_lookup_temperament_surcharge_safe
    |     assert False, f"lookup_temperament_surcharge crashed with input {s!r}: {e}"
    | AssertionError: lookup_temperament_surcharge crashed with input '': ''
    | assert False
    | Failing test case: test_lookup_temperament_surcharge_safe(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 32, in test_lookup_temperament_surcharge_safe
    |     lookup_temperament_surcharge(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\233285\Downloads\Lab 7\processor.py", line 19, in lookup_temperament_surcharge
    |     prefix = temperament_code.strip().upper()[:3]
    |              ^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\233285\Downloads\Lab 7\test_processor.py", line 34, in test_lookup_temperament_surcharge_safe
    |     assert False, f"lookup_temperament_surcharge crashed with input {s!r}: {e}"
    | AssertionError: lookup_temperament_surcharge crashed with input None: 'NoneType' object has no attribute 'strip'
    | assert False
    | Failing test case: test_lookup_temperament_surcharge_safe(
    |     s=None,
    | )
    +------------------------------------
=========================== short test summary info ===========================
FAILED test_processor.py::test_parse_walk_schedule_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_compute_pace_per_pet_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_lookup_temperament_surcharge_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================== 3 failed in 0.61s ==============================
(venv) PS C:\Users\233285\Downloads\Lab 7>
```

Corrections Made

• The schedule parser was updated to validate the input type and required fields, reject invalid or non-positive values, and handle invalid weekend codes safely.

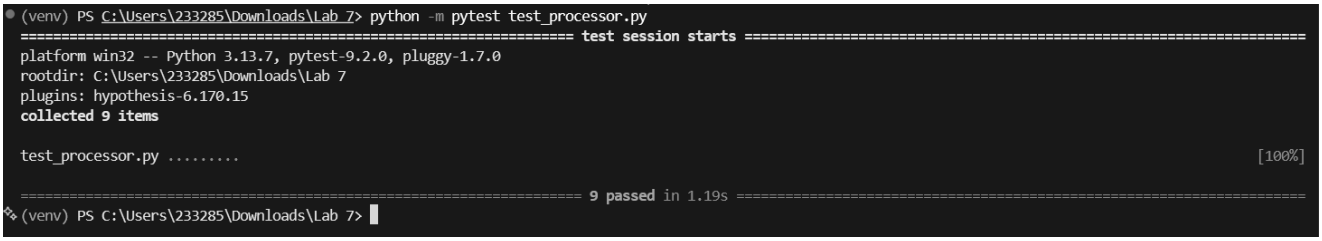
• The pace calculator was updated to validate the input format, safely convert numeric values, reject non-finite values, negative minutes and non-positive pet counts, and handle division by zero.

• The temperament lookup was updated to handle non-string and unknown inputs safely by using a safe dictionary lookup.

• A functional test exposed a prefix mismatch for PLAYFUL: the function extracts PLA, so the dictionary key was corrected from PLY to PLA.

Final Test Results

Six additional functional and validation tests were added to check expected valid outputs and invalid inputs. Together with the three Hypothesis tests, the final test suite contains nine tests.



The final test run completed successfully: 9 passed.

Conclusion

The lab demonstrated that property-based fuzz testing can uncover edge cases that may be missed by ordinary example-based testing. Hypothesis identified input-validation defects in the three functions. Defensive checks and safer lookup logic were applied, and the final test suite passed all nine tests.